# Singlet padding and the universal precession GME threshold

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**Source and statement.** Conjecture 3 of Huynh-Vu, Zaw and Scarani [1] asserts that, for every spin ensemble and every odd $K \geq 3$, a score above $T_K$ certifies genuine multipartite entanglement (GME). In units $\hbar = 1$, for a finite ensemble of positive half-integer spins $j_1, \ldots, j_N$, write

$$J_\alpha = \sum_{n=1}^{N} J_\alpha^{(n)}, \quad J_k = \cos\theta_k J_x + \sin\theta_k J_y, \quad \theta_k = \frac{2\pi k}{K}, \quad Q_K = \frac{1}{K}\sum_{k=0}^{K-1} \mathrm{pos}(J_k).$$

For a Hermitian matrix $A$, $\mathrm{pos}(A)$ has spectral weights $1, 1/2, 0$ on positive, zero, and negative eigenvalues, respectively. A density matrix $\rho \succeq 0$, $\mathrm{tr}\,\rho = 1$, has score $P_K(\rho) = \mathrm{tr}(\rho Q_K)$. A state separable across a nonempty proper subset $S$ is a convex mixture of $\rho_S \otimes \rho_{S^c}$. GME excludes convex mixtures of such states, allowing different cuts in different terms. With $c_K = 2^{-(K-1)}\binom{K-1}{(K-1)/2}$, the proposed threshold is

$$T_K = \begin{cases} 23/32, & K = 3, \\ (69 + \sqrt{181})/128, & K = 5, \\ \frac{1}{2}\left(1 + c_K \frac{K-1}{K+1}\right), & K \geq 7 \text{ odd}. \end{cases}$$

**Score-preserving padding.** Let $s = (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)/\sqrt{2}$ be the normalized two-spin-1/2 singlet. The pair operators $L_\alpha = J_\alpha^{(1)} + J_\alpha^{(2)}$ satisfy $L_x s = L_y s = 0$: applying either spin-1/2 matrix to the two terms gives cancelling coefficients. For any remaining finite ensemble, let $B_k$ be its precession observable and $A_k = L_k \otimes \mathbb{1} + \mathbb{1} \otimes B_k$ the padded observable. The isometry $V\varphi = s \otimes \varphi$ satisfies $A_k V = V B_k$. If $B_k u = \lambda u$, then $A_k V u = \lambda V u$, so $\mathrm{pos}(A_k) V u = V\,\mathrm{pos}(B_k) u$. An eigenbasis of $B_k$ therefore gives $\mathrm{pos}(A_k) V = V\,\mathrm{pos}(B_k)$, including the half weight at zero. For every remaining density matrix $\sigma$, cyclicity of the finite trace gives

$$\mathrm{tr}[(|s\rangle\langle s| \otimes \sigma)\,\mathrm{pos}(A_k)] = \mathrm{tr}[\sigma V^*\,\mathrm{pos}(A_k) V] = \mathrm{tr}[\sigma\,\mathrm{pos}(B_k)].$$

Averaging over $k$ proves $P_K(|s\rangle\langle s| \otimes \sigma) = P_K(\sigma)$. The padded state is a product across the pair–rest cut and hence is not GME.

**Consequence for unrestricted ensembles.** For fixed $K$, let $\mathcal{S}_K^{\mathrm{q}}$ be all scores from all nonempty finite positive-spin ensembles, and let $\mathcal{S}_K^{\mathrm{bisep}}$ restrict to biseparable states on ensembles with at least two particles. Padding realizes every member of $\mathcal{S}_K^{\mathrm{q}}$ in $\mathcal{S}_K^{\mathrm{bisep}}$; the reverse inclusion is immediate. Thus $\mathcal{S}_K^{\mathrm{q}} = \mathcal{S}_K^{\mathrm{bisep}}$ and their suprema agree. Any composition-independent threshold below this supremum admits a biseparable violation. This argument permits changing the ensemble; it does not identify the two optima for a fixed ensemble.

**An exact three-particle counterexample at $K = 3$.** Take physical spins $(1/2, 1/2, 3/2)$ and the pure state

$$\psi = s \otimes g, \qquad g = \frac{|3/2, 3/2\rangle - |3/2, -3/2\rangle}{\sqrt{2}}, \qquad \rho = |\psi\rangle\langle\psi|.$$

Both factors are unit vectors, so $\rho$ is positive semidefinite with trace one. Its displayed factorization proves separability across $\{1, 2\} \mid \{3\}$. All three constituent spins are strictly positive.

Here is a full finite calculation of the remaining spin's score. In the ordered $J_z$ basis with $m = 3/2, 1/2, -1/2, -3/2$, put $a = \sqrt{3}$ and

$$X = J_x^{(3/2)} = \begin{pmatrix} 0 & a/2 & 0 & 0 \\ a/2 & 0 & 1 & 0 \\ 0 & 1 & 0 & a/2 \\ 0 & 0 & a/2 & 0 \end{pmatrix}, \qquad W = \begin{pmatrix} 1 & a & a & 1 \\ a & 1 & -1 & -a \\ a & -1 & -1 & a \\ 1 & -a & a & -1 \end{pmatrix}.$$

Direct multiplication gives $W^*W = 8\mathbb{1}$ and

$$XW = W \operatorname{diag}(3/2, 1/2, -1/2, -3/2).$$

Consequently

$$\operatorname{pos}(X) = \frac{1}{8} W \operatorname{diag}(1, 1, 0, 0) W^* = \begin{pmatrix} 1/2 & a/4 & 0 & -1/4 \\ a/4 & 1/2 & 1/4 & 0 \\ 0 & 1/4 & 1/2 & a/4 \\ -1/4 & 0 & a/4 & 1/2 \end{pmatrix}.$$

For $X_\theta = \cos\theta J_x^{(3/2)} + \sin\theta J_y^{(3/2)}$, let $R_\theta = \operatorname{diag}(1, e^{i\theta}, e^{2i\theta}, e^{3i\theta})$. The standard spin matrices give $X_\theta = R_\theta X R_\theta^*$ and therefore $\operatorname{pos}(X_\theta) = R_\theta \operatorname{pos}(X) R_\theta^*$. The endpoint diagonal entries are $1/2$ and the endpoint off-diagonal entry is $-e^{-3i\theta}/4$. Since $g = (1, 0, 0, -1)^T/\sqrt{2}$,

$$\langle g| \operatorname{pos}(X_\theta) |g\rangle = \frac{1}{2} + \frac{1}{4}\cos(3\theta).$$

For each $\theta_k = 2\pi k/3$, this equals $3/4$. Padding therefore gives

$$\boxed{P_3(\rho) = \frac{3}{4} = \frac{24}{32} > \frac{23}{32} = T_3, \qquad \rho \text{ is biseparable.}}$$

This refutes Conjecture 3 with exact gap $1/32$. It also violates the implication in Result 4: $K = 3$ lies in its range, the constituent-spin sum is $5/2 \le 15$, and $1/32 > 10^{-11}$.

**What survives.** The witness lies outside Result 1's hypothesis $\sum_n j_n = K/2$; that source result is not contradicted. Nor does it contradict a bound whose hypotheses require positive effective spins on both sides of a cut. The essential distinction is that positive *constituent* spins can have an effective spin-zero sector. Omitting that sector from a maximization over cuts omits physically admissible biseparable states.

**A uniform family for every odd $K \geq 3$.** Let $L = (K-1)/2$ and take $K$ remaining spin-1/2 particles in

$$g_K = \frac{|\uparrow\rangle^{\otimes K} + (-1)^L |\downarrow\rangle^{\otimes K}}{\sqrt{2}}.$$

For $B_0 = \frac{1}{2}\sum_{n=1}^{K} \sigma_x^{(n)}$, its tensor-product $x$ eigenbasis is indexed by the number $r$ of negative local eigenvalues. The eigenvalue is $K/2 - r$, which is positive exactly when $0 \leq r \leq L$. Each eigenvector contributes $2^{-K}$ to the endpoint diagonal entries and $(-1)^r 2^{-K}$ to their off-diagonal entry. Hence those diagonal entries are $1/2$, and

$$\langle\uparrow^K| \operatorname{pos}(B_0) |\downarrow^K\rangle = 2^{-K} \sum_{r=0}^{L} (-1)^r \binom{K}{r} = (-1)^L 2^{-K} \binom{K-1}{L} = \frac{(-1)^L c_K}{2}.$$

The middle identity follows from Pascal's rule: the two alternating sums telescope, leaving $(-1)^L \binom{K-1}{L}$. Rotation about $z$ multiplies the endpoint off-diagonal entry by $e^{-iK\theta}$; for $\theta_k = 2\pi k/K$ this is one. Thus every sampled direction has the same expectation and

$$P_K(g_K) = \frac{1 + c_K}{2}, \qquad P_K(s \otimes g_K) - T_K = \begin{cases} 1/32, & K = 3, \\ (19 - \sqrt{181})/128, & K = 5, \\ c_K/(K+1), & K \geq 7 \text{ odd}. \end{cases}$$

All gaps are strictly positive: $181 < 19^2$ and $c_K > 0$. The padded state has $K + 2$ positive-spin constituents and is biseparable across the singlet–rest cut. This is an exact written proof of the family, not an inference from finitely many numerical samples.

**What the Lean development checks.** The linked module [2] defines the literal tensor-product spin observables, the spectral convention at zero, normalized density matrices, cut separability, GME, and the universal implication. Its public theorem `result : ¬ claim` proves the three-particle $K = 3$ refutation above. Its private helper `singlet_padding_ensemble_score`, used in that proof, checks the score identity for an arbitrary remaining spin list and matrix, with the literal ensemble observables. The spin parameters in Lean are doubled physical spins: `![1,1,3]` denotes $(1/2, 1/2, 3/2)$.

The all-odd GHZ formula and the score-set/supremum consequence proved here are not separate Lean theorems in this module. The merged delivery records a successful Lean build and standard axiom closure (`propext`, `Classical.choice`, `Quot.sound`) at the revision linked below; this note reports that validation, not a fresh run. Whether an appropriate exclusion of zero-spin cut sectors restores the printed threshold remains an open corrected question.

**References.**

[1] K.-N. Huynh-Vu, L. H. Zaw and V. Scarani, *Certification of genuine multipartite entanglement in spin ensembles with measurements of total angular momentum*, Phys. Rev. A 109, 042402 (2024), arXiv v2, Conjecture 3 and Result 4, p. 9. `https://arxiv.org/abs/2311.00806v2`

[2] Lean source at the recorded validated revision: `PrecessionUniversalThresholdRefutation.lean`, commit `3155010510edfab84323e848b07abac257ed5ed5`.